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import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score

# Data
years = list(range(2000, 2023))

forest_area = [
675910,679700,683490,687280,691070,694860,698650,702440,706230,710020,
713810,717600,721390,725180,728970,732760,736550,740340,744130,747920,
751710,755500,759290]

agri_land = [
1809750,1804870,1805600,1802490,1803750,1801260,1799060,1796200,1797570,
1800680,1795730,1796700,1796420,1796980,1797280,1796750,1795780,1792360,
1791160,1790450,1785280,1785423,1785567]

cattle_density = [
99.46125155,100.8382875,101.9051839,103.1905864,104.2273042,105.4817184,
106.7223995,108.00579,109.0360876,109.9584601,111.3753181,112.4283408,
113.5591899,114.6367795,115.7304371,116.8776958,118.0545501,119.3956571,
120.5922419,121.7571002,123.2299695,124.3402824,125.4503471]

# Create DataFrame
data = pd.DataFrame({
    "Forest_Area": forest_area,
    "Agricultural_Land": agri_land,
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    "Cattle_Density": cattle_density
})

# Features (Independent Variables)
X = data[["Agricultural_Land", "Cattle_Density"]]

# Target (Dependent Variable)
y = data["Forest_Area"]

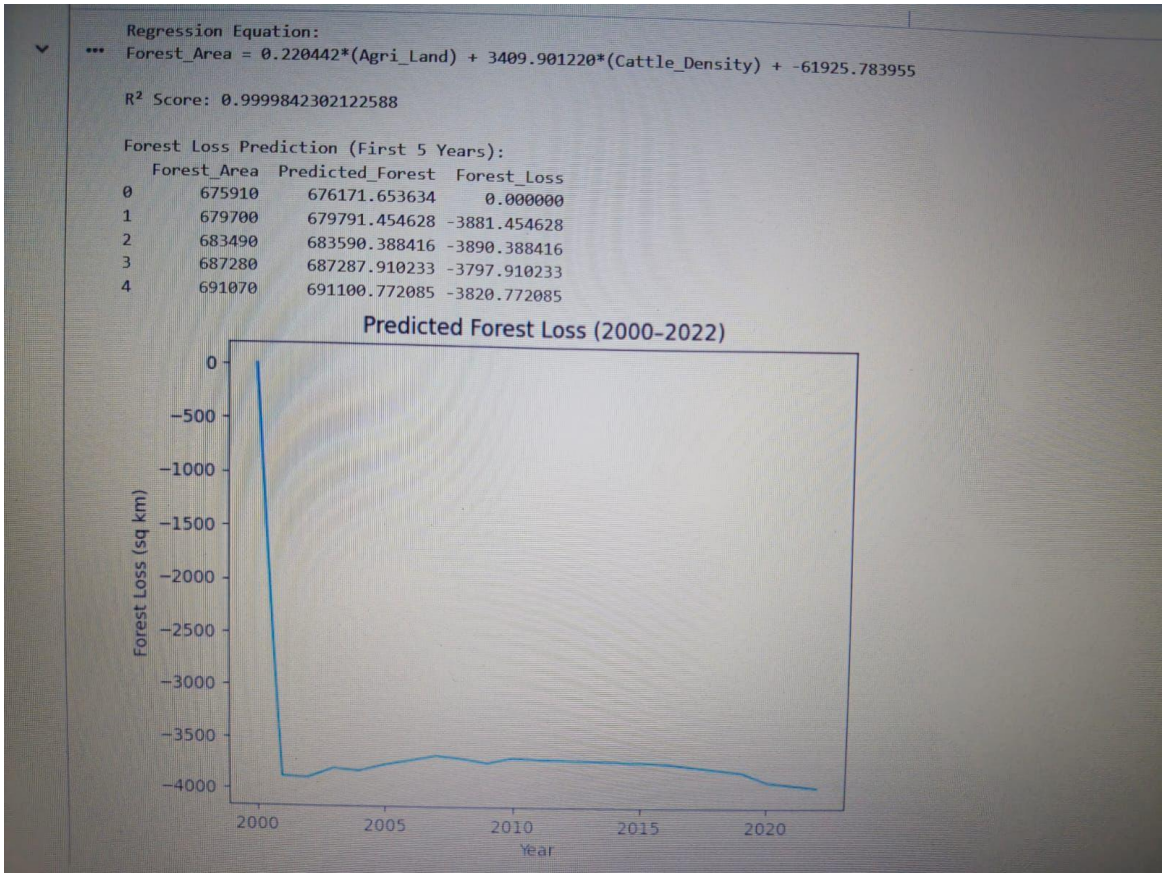
# Train Model
model = LinearRegression()
model.fit(X, y)

# Predict Forest Area
predicted_forest = model.predict(X)

# Add predictions to dataframe
data["Predicted_Forest"] = predicted_forest

# Calculate Forest Loss (if negative, then no loss)
data["Forest_Loss"] = data["Forest_Area"].shift(1) - data["Predicted_Forest"]
data["Forest_Loss"] = data["Forest_Loss"].fillna(0)

# Model Accuracy
r2 = r2_score(y, predicted_forest)

print("Regression Equation:")
print(f"Forest_Area = {model.coef_[0]:.6f}*(Agri_Land) + "
      f"{model.coef_[1]:.6f}*(Cattle_Density) + "
```

```
    f"{model.intercept_:.6f}")

print("\nR² Score:", r2)

print("\nForest Loss Prediction (First 5 Years):")
print(data[["Forest_Area", "Predicted_Forest", "Forest_Loss"]].head())

# Plot Forest Loss
plt.figure()
plt.plot(years, data["Forest_Loss"])
plt.xlabel("Year")
plt.ylabel("Forest Loss (sq km)")
plt.title("Predicted Forest Loss (2000–2022)")
plt.show()
```